

A Reproducible Benchmark of Classical Machine Learning Methods for Fault Diagnosis in the Tennessee Eastman Process

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Abstract

Fault diagnosis in industrial processes is complicated by nonlinear variable interactions, temporal dependence, and fault signatures that overlap normal operation. This work presents a reproducible benchmark of six classical machine-learning methods for multiclass diagnosis in the Tennessee Eastman Process (TEP): logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine, and XGBoost. The expanded TEP data comprised 15.33 million observations, 52 process variables, normal operation, and 20 fault classes. To control leakage, data were partitioned by complete simulation runs, and standardization parameters were estimated using training data only. Hyperparameters were selected on a validation partition, followed by repeated validation over five seeds. The prespecified primary criterion was macro-F1. XGBoost ranked first in repeated validation, with macro-F1 of 0.7683 ± 0.0017 and Matthews correlation coefficient (MCC) of 0.7426. After freezing the model and protocol, a single evaluation was performed on the complete official test partition containing 10.08 million observations. Under the literal labels distributed with the dataset, the final model achieved accuracy of 0.6754, macro-F1 of 0.7200, and MCC of 0.6621. We further show that these figures are bounded by a structural property of the benchmark itself. Because each fault is injected 160 samples into every 960-sample official test run, literal labeling imposes a recall ceiling of $800/960 = 0.8333$ on every fault class. Faults 6 and 7 attained recalls of 0.8333 and 0.8332, that is, the ceiling itself. Re-evaluating only the post-onset segment raised macro-F1 from 0.7200 to 0.8054, accuracy from 0.6754 to 0.7949, and MCC from 0.6621 to 0.7852, without refitting the model. Among the 1.6 million pre-onset test observations, 92.0% were assigned to normal operation or to the weakly observable faults 3, 9 and 15, which is the physically correct response and is nonetheless scored as error under literal labels. The effect is not specific to the selected model: seven classifiers spanning single trees, ensembles and linear baselines all show a relative accuracy gain between 0.142 and 0.150, against a structural bound of 0.1587. Rescoring the validation partition under the same protocol raised its macro-F1 from 0.7706 to 0.7953, so that the apparent validation-to-test generalization gap of +0.0483 becomes -0.0125 under a symmetric comparison. The gap therefore reflects the difference in mislabeled fraction between development runs (4%)

and test runs (16.7%) rather than a loss of generalization. Seventeen classes were diagnosed reliably, whereas normal operation and faults 3, 9, and 15 remained mutually confounded, consistent with their long-documented weak observability. Nonlinear tree ensembles provide strong diagnosis for most TEP conditions, but aggregate metrics conceal both class-specific limitations and labeling artifacts that are rarely reported.

Keywords: Tennessee Eastman Process; fault diagnosis; machine learning; XGBoost; process monitoring; reproducible benchmark; fault-onset labeling.

1 Introduction

Industrial processes operate through interacting physical, chemical, and control mechanisms. A local disturbance can propagate through feedback loops, alter several measured variables, and eventually compromise product quality, equipment integrity, safety, or environmental performance. Fault detection and diagnosis (FDD) therefore constitute central elements of modern process monitoring [1, 2, 3]. Detection determines whether the process has departed from acceptable operation, while diagnosis seeks to identify the underlying condition so that operators or automated systems can respond appropriately.

Model-based diagnosis can be effective when reliable first-principles models are available. However, plant-wide systems are frequently nonlinear, high-dimensional, stochastic, and only partially observed. These properties have motivated data-driven monitoring, in which statistical or machine-learning models infer diagnostic boundaries from historical measurements [4, 2]. Classical multivariate methods such as principal-component analysis (PCA), partial least squares (PLS), and Fisher discriminant analysis established important foundations for process monitoring [5]. More recent studies increasingly employ tree ensembles, support vector machines, neural networks, and sequence models to represent nonlinear relationships and dynamic behavior.

The Tennessee Eastman Process (TEP), originally introduced by Downs and Vogel [6], remains one of the most widely used benchmarks in this field. It represents a reactor–separator–recycle chemical process with 41 measured variables, 11 manipulated variables, and a diverse set of disturbances. Its continued value arises from the coexistence of relatively distinctive faults and difficult conditions whose trajectories overlap normal

operation. The expanded simulation data published by Rieth et al. [7] provide hundreds of independent runs per condition and non-overlapping random seeds for development and testing, enabling large-scale and reproducible evaluation.

Despite extensive use of the TEP, reported results are often difficult to compare. Studies may split temporally adjacent observations rather than complete runs, tune models on the test set, report only accuracy, or omit class-level behavior. Treating autocorrelated observations as independent experimental replicates can produce optimistic estimates, while a high global score can hide failures in specific operating conditions. These issues are especially relevant when the normal state is confused with a fault, because false alarms can reduce operator confidence and generate unnecessary interventions.

This study addresses these limitations through a leakage-controlled benchmark of six classical supervised classifiers: logistic regression, decision tree, random forest, histogram-based gradient boosting, linear support vector machine (SVM), and XGBoost. The study asks five questions:

1. Which classical model provides the strongest macro-averaged diagnostic performance under repeated validation?
2. Does the selected model maintain its performance on a complete, independent official test partition?
3. Which operating conditions account for the remaining diagnostic errors?
4. How much of the reported error is attributable to the fault-onset labeling convention rather than to the classifier?
5. Which model offers the best compromise between diagnostic performance and computational cost?

The main contribution is an auditable protocol that separates model selection from final testing, preserves complete simulation runs, quantifies variability over five seeds, reports confidence intervals, and examines precision, recall, F1, and confusion patterns for all 21 classes. Rather than presenting a single headline score, the analysis identifies a compact ambiguity group that defines the practical limit of the selected classifier.

A second contribution is methodological and, in our view, of wider consequence for the TEP literature. The expanded dataset injects each fault after a fixed warm-up interval, so a fraction of every faulty run is nominally normal yet carries the fault label. This fraction differs between development runs (4%) and official test runs (16.7%). We quantify how much of the reported error and of the apparent generalization gap is produced by this convention alone, and we report every headline metric under both the literal labeling and a post-onset labeling. To our knowledge this decomposition is not routinely reported, which may account for part of the dispersion among published TEP results.

2 Related Work

2.1 Data-driven process monitoring

Data-driven FDD spans latent-variable modeling, classification, clustering, change detection, and temporal modeling. The review by Venkatasubramanian et al. [1] established a broad taxonomy covering quantitative models, qualitative models, and process-history-based methods. Chiang et al. [5, 4] showed how PCA, discriminant PLS, and Fisher discriminant analysis could support chemical-process diagnosis. Later reviews emphasized that industrial data may be nonlinear, dynamic, nonstationary, multimodal, and autocorrelated [2, 3]. These properties motivate flexible nonlinear classifiers, but also demand careful validation.

A substantial line of work addresses the serial correlation of process data directly. Dynamic PCA augments the data matrix with time-lagged variables to capture time-varying structure [26]; canonical variate analysis models the dynamics explicitly and was compared against PCA and DPCA on the TEP simulator by Russell et al. [27], later extended with kernel density estimation for nonlinear dynamics [28]. These methods are relevant here for a reason beyond their performance: they treat the temporal structure of a run as the object of analysis, whereas an observation-level classifier treats each row independently. The labeling question examined in Section 3.2 arises precisely at that boundary.

2.2 Classical machine learning for TEP diagnosis

Decision trees recursively partition the feature space into regions associated with class labels [8]. Random forests reduce the instability of individual trees by combining bootstrap samples and randomized feature subsets [9]. Gradient boosting instead constructs an additive ensemble in which successive learners correct residual errors [10]. XGBoost extends this principle with regularized objectives, efficient histogram-oriented computation, column subsampling, and systems-level optimizations [11]. These characteristics make tree ensembles attractive for TEP data, where variables interact nonlinearly.

Linear models provide important controls. Multinomial logistic regression tests whether approximately linear decision surfaces are sufficient, whereas linear SVMs maximize class separation through margin-based learning [12]. Their inclusion helps determine whether the added complexity of nonlinear ensembles is justified. All implementations in this study were based on scikit-learn [13], except XGBoost, which used its reference implementation [11].

Deep architectures have been applied extensively to the same benchmark. Chadha and Schwung compared feedforward and convolutional networks for TEP fault detection [22]; hybrid designs combining support vector machines with latent-variable preprocessing [25], hierarchical LSTM structures [23], and metaheuristically optimized deep models [24] have all reported strong results. Most directly comparable to the present study, Lomov et al. [21] evaluated recurrent, attention-based

and temporal convolutional architectures on the same expanded TEP dataset used here. The present work does not propose a competing architecture; it examines a property of the evaluation protocol that applies to all of them.

2.3 Evaluation protocols and leakage

Concern about evaluation protocol is not specific to process monitoring. In a survey across disciplines that have adopted machine learning, Kapoor and Narayanan identified leakage in seventeen fields, affecting at least 294 publications and in several cases producing conclusions that did not survive correction [20]. Their taxonomy distinguishes eight mechanisms, two of which bear directly on the present benchmark: temporal leakage, where information from the future informs a prediction about the past, and non-independence between training and evaluation samples, where records that share a common origin are distributed across partitions and treated as independent replicates.

Both are latent in observation-level TEP classification. Consecutive samples within a simulation run are strongly autocorrelated, so a split performed at the observation level places near-duplicate records on both sides of the partition. The run-level protocol of Section 3.3 is designed to avoid this. The labeling question analyzed in this paper is related but distinct: it concerns not how records are divided, but whether the label attached to a record describes the state of the plant at that instant.

2.4 Evaluation beyond accuracy

Accuracy can be misleading when performance varies substantially among classes. Macro-F1 gives equal importance to every class by averaging class-specific F1 scores, while weighted-F1 weights them by support. Balanced accuracy averages class recalls. MCC summarizes agreement between predictions and true labels and remains informative for multiclass problems [14]. Confusion matrices are indispensable because they reveal whether errors are diffuse or concentrated in specific class pairs. The present study therefore uses macro-F1 as the primary model-selection criterion and treats the other metrics as complementary evidence.

2.5 Reported treatment of the fault-onset window

The analysis reported in this paper is only consequential if the treatment of the onset window varies, or goes unstated, across published work on the same benchmark. We therefore examined the data-description and experimental-protocol sections of representative observation-level studies, recording for each one the dataset variant, the unit of partitioning, and whether the handling of the pre-onset segment is stated explicitly. Table 1 summarizes the outcome.

The question applies to both distributions of the benchmark. In the original data of Downs and Vogel as distributed by Chiang et al. [6, 4], each condition

is simulated for 24 hours in training and 48 hours in testing at a three-minute interval, giving 480 and 960 samples, with the fault injected after one hour and eight hours respectively. The pre-onset fractions are therefore 4.2% and 16.7%, essentially identical to those of the expanded dataset analyzed here. The artifact is thus a property of the benchmark design rather than of any particular redistribution of it, and the results of Section 4.4 bound what can be reported on either version.

One case illustrates the pattern and the stakes. Agarwal et al. [23] describe their data as 1440 samples per class, of which the first 480 are used for calibration and the remaining 960 for testing, and identify the source distribution explicitly. The instant of fault injection is not mentioned anywhere in the data description or the experimental protocol, so all 960 test samples are scored against the fault label although the first 160 precede injection. The consequence is not incidental to that study’s argument: its stated objective is to improve the diagnosis of the incipient faults 3, 9 and 15, to which end pseudo-random binary signals are actively injected into the plant. Under literal labeling, part of the difficulty that motivates such an intervention is contributed by observations in which no fault is yet present.

[TBD: Complete the remaining rows and summarize: how many of the examined studies state the treatment of the onset window, how many partition by run rather than by observation, and whether the reported figures are consequently comparable.]

3 Materials and Methods

3.1 Tennessee Eastman Process data

The TEP simulates a plant-wide reactor–separator–recycle system and includes 41 process measurements (XMEAS) and 11 manipulated variables (XMV), totaling 52 predictors [6]. The classification task contained 21 labels: class 0 represented fault-free operation and classes 1–20 represented the predefined fault conditions. The CSV distribution used in this work is a format conversion of the expanded TEP simulation data of Rieth et al. [7, 15].

The four source files contained 15,330,000 observations and 55 columns: the 52 process variables plus the identifiers `faultNumber`, `simulationRun`, and `sample`. Development runs contained 500 observations each. Official test runs contained 960 observations each. Every class had 500 official test runs, yielding

$$21 \times 500 \times 960 = 10,080,000 \quad (1)$$

test observations. Each class therefore contributed 480,000 observations to the final test.

3.2 Fault-onset windows and labeling protocols

The expanded dataset is generated by simulating each condition for 25 hours (development) or 48 hours (official

Table 1: Reported experimental protocol in representative observation-level studies on the Tennessee Eastman benchmark. “Onset stated” records whether the treatment of observations preceding fault injection is described explicitly.

Study	Methods	Dataset	Split unit	Onset stated	Metric
Yin et al. [18]	PCA, PLS, FDA, ICA	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Gao & Hou [25]	GS-PCA + SVM	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Chadha & Schwung [22]	DNN, CNN	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Lomov et al. [21]	LSTM, GRU, TCN	[TBD:]	[TBD:]	[TBD:]	[TBD:]
Agarwal et al. [23]	Hierarchical LSTM-SAE	original	simulation file	no	FDR, FAR
Anitha et al. [24]	Hybrid DL	[TBD:]	[TBD:]	[TBD:]	[TBD:]
This work	Six classical classifiers	expanded	complete run	yes, both	macro-F1, MCC

test) with a sampling interval of three minutes, and by injecting the fault after a fixed warm-up interval: one hour in faulty development runs and eight hours in faulty test runs [7, 16, 17]. In sample indices, the fault is active from sample 21 in development runs and from sample 161 in official test runs. Fault-free runs contain no injection.

Consequently, the labels distributed with the dataset are only conditionally correct. In a faulty development run, samples 1–20 (4.0% of the run) describe nominal operation while carrying a fault label; in a faulty official test run, samples 1–160 (16.7% of the run) do so. Two consequences follow directly and are, to our knowledge, seldom made explicit.

First, under an observation-level protocol with literal labels, the recall of every fault class is bounded above by

$$R_c^{\max} = \frac{960 - 160}{960} = 0.8333, \quad c = 1, \dots, 20, \quad (2)$$

whenever the classifier assigns pre-onset observations to some class other than c . No such ceiling applies to class 0, whose test runs contain no injection. Second, because the mislabeled fraction differs between development (4.0%) and test (16.7%), a metric computed on the two partitions is not measuring the same quantity, and part of any observed validation-to-test degradation is attributable to this asymmetry rather than to generalization.

We therefore report all headline metrics under two labeling protocols:

- **Literal labeling (L):** every observation retains the label distributed with the dataset. This is the protocol implicitly adopted by most published observation-level TEP results and is retained here for comparability.
- **Post-onset labeling (P):** pre-onset observations of faulty runs are excluded *at evaluation time*, so that each scored observation carries a label consistent with the physical state of the plant. The model is not refitted: protocol P rescores the predictions of the same frozen model that was fitted under protocol L, including its 4% of contaminated training observations. The comparison L versus P therefore isolates the labeling convention alone,

and the reported post-onset figures are conservative, since a model additionally trained without the contaminated observations could only do better.

A third variant, in which pre-onset observations are excluded from fitting as well, was not evaluated here; it would confound the labeling effect with a change in the training distribution and is left for future work.

Protocol L is used for model selection, so that the frozen-model decision is unaffected by the present analysis. Protocol P is reported alongside it in Section 4.4. A third option, relabeling pre-onset observations as class 0, was not adopted because it alters the class prior of the normal condition and would confound the analysis of the ambiguity group.

3.3 Leakage-controlled partitioning and preprocessing

The original development runs were divided at the run level into training and validation partitions using a fixed random seed and a stratified 80/20 split within each source and class. This produced 8,400 training runs (400 per class), 2,100 validation runs (100 per class), and 10,500 official test runs (500 per class). No run was assigned to more than one partition.

All 52 predictors were standardized using

$$z_{ij} = \frac{x_{ij} - \mu_j^{(\text{train})}}{\sigma_j^{(\text{train})}}, \quad (3)$$

where $\mu_j^{(\text{train})}$ and $\sigma_j^{(\text{train})}$ were estimated exclusively from the 4.2 million training observations. The fitted transformation was then applied unchanged to validation and test data. This procedure prevented information from the validation or test partitions from influencing preprocessing.

3.4 Candidate classifiers

Six model families were compared. Hyperparameter tuning evaluated three candidates per family, producing 18 configurations. The selected configurations were subsequently frozen:

- **Logistic regression:** $C = 1.0$, maximum 1,000 iterations;

- **Decision tree:** unrestricted depth and minimum leaf size 1;
- **Random forest:** 400 trees, unrestricted depth, 50% of features considered per split, and minimum leaf size 1;
- **HistGradientBoosting:** learning rate 0.05, 350 boosting iterations, and at most 31 leaf nodes;
- **Linear SVM:** stochastic-gradient hinge-loss classifier with $\alpha = 10^{-5}$, maximum 2,000 iterations, and tolerance 10^{-3} ;
- **XGBoost:** 250 trees, maximum depth 6, learning rate 0.1, row subsampling 0.8, column subsampling 0.8, multiclass probabilistic objective, and histogram tree construction.

The tuning stage ranked configurations by macro-F1 and used MCC as a secondary criterion. The official test partition was not accepted as an input by either the tuning or repeated-validation programs.

The linear SVM was implemented as a stochastic-gradient surrogate rather than as an exact margin solver, because quadratic-programming solvers were not tractable at this sample size. Its score should therefore be read as a lower bound on linear-margin performance, and the evidence for nonlinearity reported in Section 5.1 rests primarily on the multinomial logistic regression, which was optimized to convergence. Kernel SVMs were not evaluated for the same computational reason.

3.5 Repeated validation

After selecting the best configuration for each model family, variability was measured across seeds 2026–2030. For each seed, 40 complete runs per class were sampled from training and independently from validation. Thus, every repetition used 420,000 training and 420,000 validation observations while retaining the full 500-observation trajectories of all selected runs.

For each metric, the mean, sample standard deviation, and 95% confidence interval were calculated across the five repetitions. With $n = 5$, the interval was

$$\bar{x} \pm t_{0.975,4} \frac{s}{\sqrt{5}}. \quad (4)$$

Because only five seeds were used, these intervals quantify observed seed-to-seed variability but should not be interpreted as definitive evidence of pairwise statistical superiority. In particular, overlapping intervals for the two leading models were interpreted conservatively.

In addition to the interval estimates, the six model families were compared with the Friedman rank test over the five seeds treated as blocks, followed by the Nemenyi post-hoc procedure [19]. Pairwise comparisons used the Wilcoxon signed-rank test with Holm correction. The critical difference of the Nemenyi procedure was computed as $CD = q_\alpha \sqrt{k(k+1)/6N}$ with $k = 6$ models and $N = 5$ blocks. Results are reported in Section 4.1. With five blocks these procedures have low power by construction, and they are reported to document that

the prespecified ranking rule was not substituted for an inferential check, not to establish pairwise superiority.

Because a single frozen evaluation cannot express sampling uncertainty, the official-test metrics were additionally accompanied by 95% intervals obtained by resampling complete runs with replacement (1,000 bootstrap replicates over the 10,500 test runs). The run, not the observation, is the resampling unit, consistent with the argument that temporally adjacent observations are not independent replicates.

3.6 Frozen final evaluation

XGBoost candidate 1 was selected because it achieved the highest repeated-validation macro-F1. The final seed was fixed at 2026 before test evaluation. The final development set combined 40 complete runs per class from training and 40 from validation, totaling 840,000 observations. The frozen model was then evaluated once on all 10,080,000 official test observations, without subsampling or post-test adjustment.

Two earlier executions aborted during structural validation, before any model fitting or metric computation, because of an incorrect run-length assertion; only that assertion was changed, and the full incident log is available in the repository.

The choice of 840,000 development observations, rather than the full 5,250,000 available, was imposed by the computational budget. To verify that this choice does not materially affect the conclusions, a learning curve was computed on the validation partition over $\{10, 20, 40, 80, 160\}$ complete runs per class: **[TBD: learning-curve result and saturation point]**.

3.7 Performance measures and reproducibility

The reported measures were accuracy, balanced accuracy, precision, recall, class-specific F1, macro-F1, weighted-F1, MCC, confusion matrix, training time, inference time, inference time per observation, and serialized model size. Zero divisions in class-specific measures were assigned zero. Data and output integrity were monitored with SHA-256 hashes and manifests. The final execution used Python 3.12.13, scikit-learn 1.9.0, and XGBoost 3.4.0 on Linux.

4 Results

4.1 Model selection under repeated validation

Table 2 summarizes the five-seed validation results. XGBoost ranked first with macro-F1 of 0.7683 ± 0.0017 and MCC of 0.7426 ± 0.0018 . HistGradientBoosting was a close second, with macro-F1 of 0.7643 ± 0.0034 . Random forest achieved 0.7398, followed by decision tree at 0.6714. The two linear baselines performed substantially worse: logistic regression achieved 0.4685 and linear SVM achieved 0.4438.

Table 2: Repeated-validation performance over five seeds. Macro-F1 was the prespecified primary selection criterion.

Rank	Model	Macro-F1 (mean $\pm$ SD)	95% CI	MCC (mean $\pm$ SD)
1	XGBoost	0.7683 $\pm$ 0.0017	[0.7661, 0.7705]	0.7426 $\pm$ 0.0018
2	HistGradientBoosting	0.7643 $\pm$ 0.0034	[0.7601, 0.7686]	0.7391 $\pm$ 0.0045
3	Random Forest	0.7398 $\pm$ 0.0021	[0.7372, 0.7424]	0.7159 $\pm$ 0.0021
4	Decision Tree	0.6714 $\pm$ 0.0013	[0.6699, 0.6730]	0.6537 $\pm$ 0.0015
5	Logistic Regression	0.4685 $\pm$ 0.0074	[0.4592, 0.4777]	0.4567 $\pm$ 0.0073
6	Linear SVM	0.4438 $\pm$ 0.0099	[0.4314, 0.4561]	0.4377 $\pm$ 0.0109

The Friedman test over the five seeds rejected the hypothesis of equal performance among the six families ($\chi^2_5 = 24.54$, $p = 1.71 \times 10^{-4}$), with a Kendall coefficient of concordance of $W = 0.982$. The ordering is thus almost perfectly consistent from seed to seed: XGBoost obtained the best macro-F1 in four of the five repetitions, and the mean ranks were 1.2 (XGBoost), 1.8 (HistGradientBoosting), 3.0 (random forest), 4.0 (decision tree), 5.0 (logistic regression) and 6.0 (linear SVM).

Post-hoc resolution is nevertheless limited. With $N = 5$ blocks the Nemenyi critical difference is 3.372 mean ranks, so only three of the fifteen pairwise comparisons are individually significant, all of them between a tree ensemble and a linear baseline. The separation between XGBoost and HistGradientBoosting is 0.6 mean ranks, far below the critical difference. The Wilcoxon signed-rank test cannot help here either: with five paired observations its smallest attainable two-sided p -value is 0.0625, so no comparison can reach $\alpha = 0.05$ even under a perfect win record, and after Holm correction none does.

These two facts are complementary rather than contradictory. The high concordance shows that the ranking is not an artifact of seed choice; the wide critical difference shows that five repetitions cannot certify small differences. Both support the conservative reading adopted here: the ordering is reproducible, the gap between the two leading models is not resolvable at this sample size, and additional seeds would be required to settle it.

Figure 1 shows the ranking and its uncertainty. The confidence intervals of XGBoost and HistGradientBoosting overlapped. Accordingly, XGBoost was selected by the prespecified macro-F1 rule, but the result is not presented as proof of a statistically significant difference between the two models. The large separation between nonlinear ensembles and linear baselines indicates that nonlinear interactions are central to this diagnostic task.

4.2 Official test performance

After the protocol was frozen, the selected XGBoost model was fitted to 840,000 development observations and evaluated on the complete official test set. Table 3 presents the final metrics under literal labeling. Accuracy and balanced accuracy were both 0.6754, macro-F1 and weighted-F1 were both 0.7200, and MCC was 0.6621. The equality of accuracy and balanced accuracy, and of macro-F1 and weighted-F1, follows from the exactly balanced class supports, and holds only under literal la-

beling; under the post-onset protocol of Section 4.4 the fault classes retain 400,000 observations while the fault-free class retains 480,000, so the two pairs of metrics separate. Macro-F1 exceeds accuracy because accuracy equals macro-recall under balanced supports, while precision exceeds recall in most classes; the harmonic mean of a high precision and a moderate recall is larger than the recall alone. The gap between macro-F1 and accuracy is therefore itself a signature of the systematic under-recall documented in Section 4.4.

Table 3: Frozen XGBoost performance on the official test set.

Metric	Value
Accuracy	0.6754
Balanced accuracy	0.6754
Macro-F1	0.7200
Weighted-F1	0.7200
MCC	0.6621
Training time (s)	170.34
Inference time (s)	56.57
Inference (ms/observation)	0.005612
Development observations	840,000
Test observations	10,080,000

Relative to repeated validation, macro-F1 decreased by 0.0483 and MCC by 0.0805. Three mechanisms could account for this difference: independent random states in the test simulations, the longer duration of test runs, and the larger fraction of mislabeled pre-onset observations. The third mechanism is both the largest and the only one that is purely a labeling artifact. Under literal labeling the attainable recall per fault class falls from $480/500 = 0.96$ in development runs to $800/960 = 0.8333$ in test runs, a ratio of 0.868, which is of the same order as the observed degradation. Section 4.4 shows that the gap does not merely shrink under post-onset labeling but changes sign: the post-onset test macro-F1 of 0.8054 exceeds the literal-label validation figure of 0.7683 by 0.0371.

That comparison alone would not be symmetric, because the validation figure was itself computed under literal labels and carries its own 4% pre-onset contamination. The validation partition was therefore rescored under both protocols as well. Applying the selected XGBoost configuration to all 1,050,000 validation observations gave a literal-label macro-F1 of 0.7706, which agrees with the five-seed repeated-validation figure of 0.7683 to within 0.0023 and confirms that this partition is a faithful reference. Excluding the 40,000 pre-onset

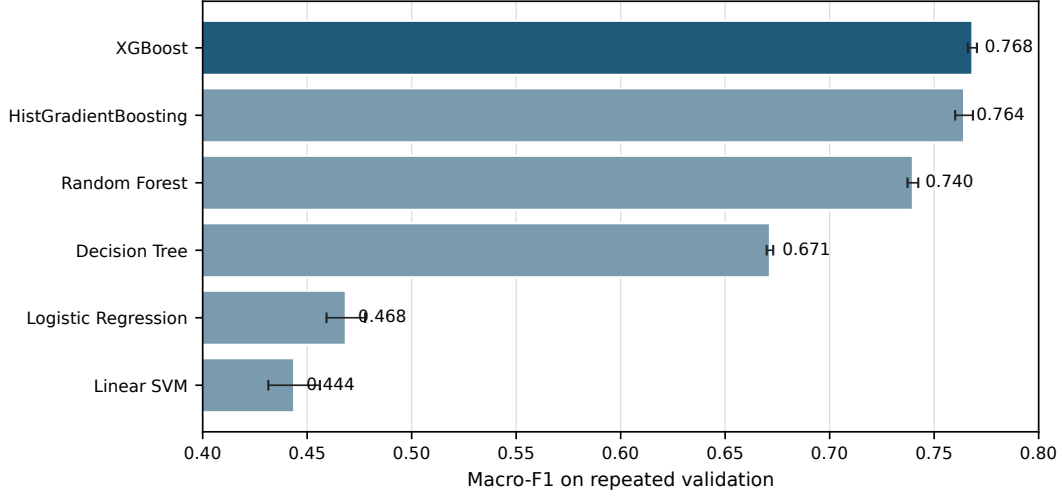


Figure 1: Repeated-validation macro-F1 for the six candidate algorithms. Error bars denote 95% confidence intervals across five seeds.

observations, 3.81% of the partition, raised macro-F1 to 0.7953, accuracy from 0.7556 to 0.7843, and MCC from 0.7440 to 0.7739.

The correction on the validation partition is thus $+0.0246$ in macro-F1, against $+0.0854$ on the test partition, a ratio of 0.29 that tracks the ratio of contaminated fractions (3.81% against 15.87%). Transferring the measured validation correction to the repeated-validation reference gives a post-onset validation macro-F1 of approximately 0.7929, against a post-onset test macro-F1 of 0.8054. Table 4 summarizes the decomposition.

Table 4: Validation-to-test difference in macro-F1 under both labeling protocols. The gap present under literal labeling does not survive a symmetric comparison.

Protocol	Validation	Test	Difference
Literal (L)	0.7683	0.7200	$+0.0483$
Post-onset (P)	0.7929	0.8054	-0.0125

Under a symmetric comparison the gap is not merely reduced but reversed, and its magnitude falls to roughly a quarter of the literal-label value, within the range of the seed-to-seed variability of Table 2. We therefore conclude that the reported generalization gap is an artifact of the labeling convention rather than evidence of degradation on unseen trajectories, and that the literal-label figure of 0.7200 understates the discriminative capacity of the frozen model by 0.0854 in macro-F1.

Two caveats attach to this conclusion. The validation rescaling used the tuned configuration fitted to the full training partition, not the five per-seed refits of the repeated-validation procedure, so the transferred correction is an estimate of the effect rather than an exact recomputation; the close agreement between 0.7706 and 0.7683 bounds the error of that substitution. Furthermore, the pre-onset destinations differ between partitions: 61.1% of pre-onset validation observations were assigned to the group $\{0, 3, 9, 15\}$ against 92.0% in the test partition, which is consistent with the development segment being only 20 samples long and overlapping

the simulation transient.

4.3 Class-specific performance

Figure 2 and Table 5 demonstrate marked heterogeneity. Seventeen classes achieved F1 scores from 0.7429 to 0.9080. Classes 6 and 7 were strongest, with F1 of 0.9080 and 0.9075, respectively. Classes 1, 2, 5, and 14 also approached or exceeded 0.8958.

Four classes were persistently difficult: class 9 ($F1=0.1635$), class 15 (0.1683), normal operation (class 0; 0.1718), and class 3 (0.2357). Class 3 had the highest recall within this group (0.4218) but very low precision (0.1636), indicating systematic overprediction.

A pattern in the recall column deserves emphasis. Classes 6 and 7 reached recalls of 0.8333 and 0.8332, which coincide with the structural ceiling of Eq. (2) to within one part in ten thousand. Classes 1, 2, 4, 5 and 14 lie between 0.8187 and 0.8243, that is, at 98–99% of that ceiling. The frozen model is therefore essentially saturated on these classes once the pre-onset segment is removed, and their reported F1 values of roughly 0.88–0.91 are limited by the labeling convention rather than by the classifier.

4.4 Effect of the fault-onset window

Table 6 compares the frozen model under the two labeling protocols of Section 3.2. No refitting, retuning or reselection was performed; the same frozen model and the same predictions are scored under two label conventions.

Excluding the 1,600,000 pre-onset observations, 15.87% of the partition, raised macro-F1 by 0.0854, accuracy by 0.1195 and MCC by 0.1231. No model was refitted and no hyperparameter was changed.

Two diagnostics establish that this is a labeling effect rather than a favourable choice of subset. First, the frozen model assigned 16.10% of the pre-onset observations to class 0 and 75.86% to classes 3, 9 or 15, that is, 92.0% to the group of conditions that are indistinguish-

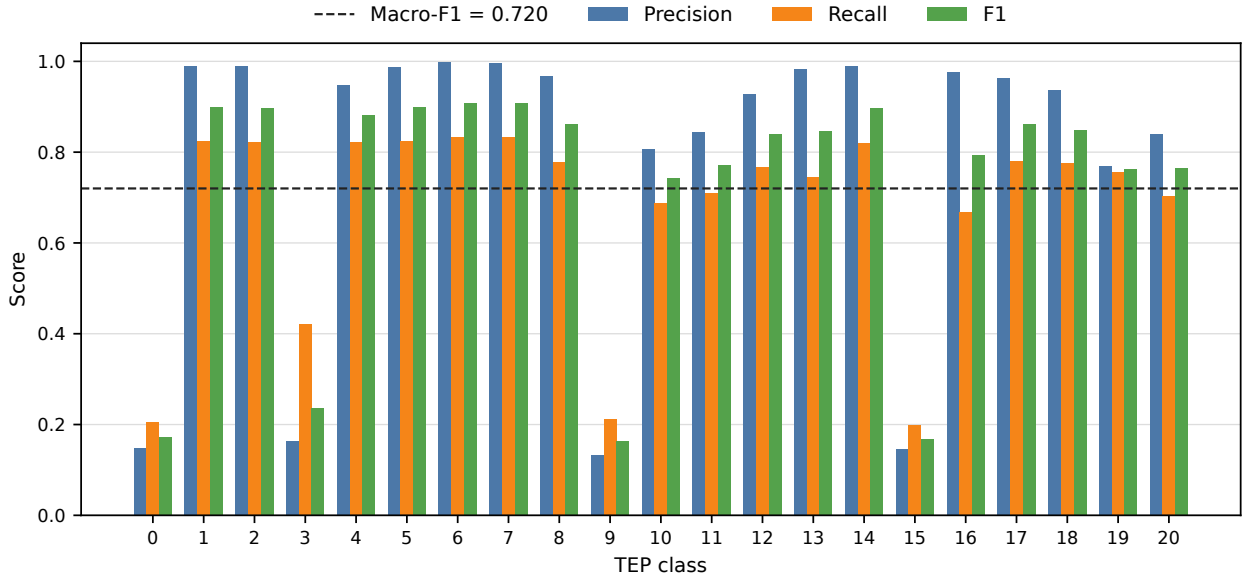


Figure 2: Class-wise precision, recall, and F1 on the complete official test partition. The dashed line indicates overall macro-F1. Class 0 is normal operation.

Table 5: Class-wise results for the frozen XGBoost model on the official test set. Every class had support of 480,000 observations.

Class	Precision	Recall	F1	Class	Precision	Recall	F1
0	0.1481	0.2045	0.1718	11	0.8447	0.7095	0.7712
1	0.9894	0.8243	0.8993	12	0.9271	0.7669	0.8395
2	0.9896	0.8209	0.8974	13	0.9828	0.7444	0.8471
3	0.1636	0.4218	0.2357	14	0.9889	0.8187	0.8958
4	0.9477	0.8222	0.8805	15	0.1465	0.1978	0.1683
5	0.9879	0.8234	0.8982	16	0.9756	0.6681	0.7931
6	0.9973	0.8333	0.9080	17	0.9624	0.7809	0.8622
7	0.9964	0.8332	0.9075	18	0.9367	0.7760	0.8488
8	0.9665	0.7779	0.8620	19	0.7701	0.7559	0.7630
9	0.1328	0.2126	0.1635	20	0.8391	0.7029	0.7650
10	0.8066	0.6885	0.7429				

able from nominal operation, and only 4.20% to the fault label they nominally carry. The model therefore identifies the pre-onset segment as a period in which nothing detectable is occurring, which is correct, and literal labeling scores this as error. Second, restricting the evaluation to post-onset observations raised the recall of the 17 well-separated classes to between 0.8015 and 0.9999, the upper value corresponding to essentially complete recovery of faults 6 and 7, while the ambiguity group improved only marginally (F1 of 0.2217, 0.3238, 0.2100 and 0.2106 for classes 0, 3, 9 and 15, against 0.1718, 0.2357, 0.1635 and 0.1683 under literal labels). The ambiguity group is therefore not a labeling artifact; the ceiling on the remaining seventeen classes is.

The two phenomena are nevertheless connected. Class 3 was predicted 1,237,562 times against a true support of 480,000, an excess of 757,562 observations. A substantial part of that excess consists of pre-onset segments belonging to other faults, which the model routes to the weakly observable conditions because, at those instants, nothing has yet happened. The overprediction of class 3 is thus partly a consequence of the labeling convention and partly a consequence of genuine overlap, and only

the post-onset analysis separates the two.

This distinction matters for comparability. As documented in Section 2.5, published observation-level results on the expanded TEP dataset do not consistently state whether pre-onset samples were retained, discarded, or relabeled, yet the three choices are separated by several points of macro-F1 for an identical model. Two studies reporting different scores may therefore differ in labeling convention rather than in method. We recommend that the treatment of the onset window be reported explicitly as part of the experimental protocol, in the same way that the unit of partitioning is reported.

4.5 Generality of the labeling effect across model families

If the effect of Section 4.4 is a property of the labeling convention rather than of the selected classifier, it should appear in every model evaluated on the same partition. To test this, an auxiliary experiment was rescored under both protocols. That experiment pre-dates the frozen benchmark, uses a per-class subsample

Table 6: Frozen XGBoost under literal (L) and post-onset (P) labeling on the official test partition. Intervals are 95% run-level bootstrap intervals over 1,000 replicates.

Metric	Literal (L)	Post-onset (P)
Accuracy	0.6754 [0.6716, 0.6797]	0.7949 [0.7898, 0.8006]
Balanced accuracy	0.6754 [0.6748, 0.6760]	0.8006 [0.7998, 0.8013]
Macro-F1	0.7200 [0.7192, 0.7208]	0.8054 [0.8044, 0.8063]
MCC	0.6621 [0.6579, 0.6666]	0.7852 [0.7798, 0.7910]
Observations	10,080,000	8,480,000

of the test partition, and includes an extremely randomized trees model in place of XGBoost; its absolute levels are therefore not comparable with Table 3, and only the differences between protocols are interpreted. No model was refitted.

The five tree-based models gained between 0.0821 and 0.0854 in macro-F1, a spread of 0.003 across estimators whose capacities differ by orders of magnitude, from a single unpruned tree to a regularized boosting ensemble. No overfitting mechanism produces that uniformity, because the degradation associated with overfitting scales with model capacity.

The relative accuracy gain explains the pattern quantitatively. If a classifier assigns pre-onset observations to any class other than the labeled fault, its literal accuracy is its post-onset accuracy scaled by the retained fraction of the partition, so that $\Delta A/A_P$ is bounded above by 0.1587 independently of the model. The seven models fall between 0.142 and 0.150, the residual being the small fraction of pre-onset observations that match the literal label by chance. The linear baselines gain less in absolute terms, 0.0415 and 0.0448, simply because they start lower; in relative terms they are affected almost identically. They also route far fewer pre-onset observations to the group $\{0, 3, 9, 15\}$, 20.6% and 41.9% against 65.7–70.2% for the ensembles, and are penalized in the same proportion regardless.

The consequence is that the effect is not a property of any estimator. Any observation-level result reported on this dataset under literal labels understates the post-onset discriminative capacity of the model by a factor predictable from its own accuracy. Comparisons between published results are correspondingly affected whenever the treatment of the onset window differs or is unstated.

4.6 Structure of the remaining errors

The normalized confusion matrix in Fig. 3 shows that errors were not uniformly distributed. They concentrated in the four-class group $\{0, 3, 9, 15\}$. Among true class-9 observations, 32.39% were predicted as class 3, 20.24% as class 0, and 18.08% as class 15. Normal observations were assigned to classes 3, 9, and 15 in 32.02%, 21.51%, and 18.24% of cases, respectively. For class 15, the corresponding proportions were 30.96%, 20.53%, and 20.03% for predicted classes 3, 9, and 0.

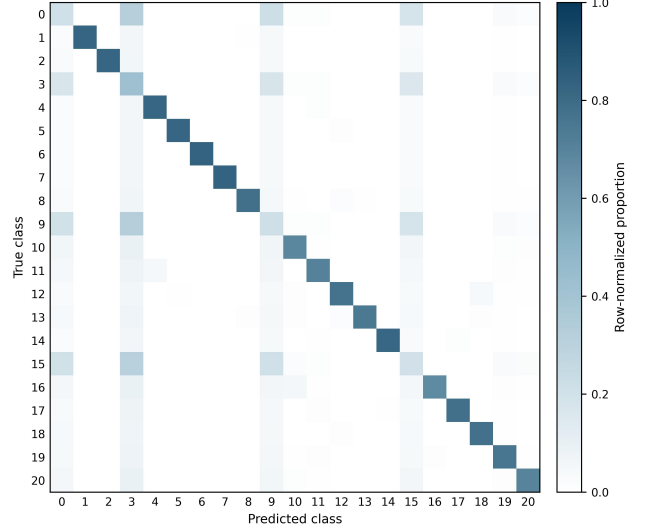


Figure 3: Row-normalized confusion matrix for the frozen XGBoost model on the official test partition.

Class 3 was predicted 1,237,562 times despite having 480,000 true observations. This excess explains its low precision and demonstrates that the model used class 3 as a frequent destination for ambiguous trajectories. The persistence of the same difficult group during repeated validation and official testing suggests an intrinsic overlap under an observation-level representation rather than an isolated sampling artifact.

4.7 Computational performance

Table 8 reports the cost of all six families under identical conditions during repeated validation, which is required to answer the performance–cost question and cannot be inferred from the selected model alone. Training times are reported as means over the five seeds and are not directly comparable with the 170.34s of Table 3, which was measured during the final evaluation on twice as many observations. The degree of parallelism was an invocation-time argument of each stage rather than a fixed setting, and it was not recorded in the run manifests, so the absolute times of the two stages reflect different thread counts on a 24-core host. The comparison between families within Table 8 is unaffected, because those six runs shared a single invocation, but the cross-stage comparison is not meaningful and no conclusion is drawn from it. Recording the degree of parallelism alongside the timing is a reproducibility requirement that this benchmark did not initially meet, and it has been added to the released code.

Table 7: Effect of post-onset labeling across model families. The last column reports the relative accuracy gain $\Delta A/A_P$, whose structural upper bound is $1 - 8,480,000/10,080,000 = 0.1587$.

Model	Macro-F1 (L)	Macro-F1 (P)	$\Delta A/A_P$
Decision tree	0.6330	0.7167	0.148
Extra trees	0.6965	0.7786	0.148
Random forest	0.7081	0.7912	0.149
HistGradientBoosting	0.7128	0.7972	0.149
XGBoost [†]	0.7200	0.8054	0.150
Logistic regression	0.4283	0.4698	0.142
Linear SVM	0.4207	0.4655	0.142

[†]Frozen model of Table 3, complete test partition.

Table 8: Performance and computational cost of the six families under repeated validation (mean over five seeds, 420,000 training observations per repetition). Times measured on the environment of Section 3.7.

Model	Macro-F1	MCC	Train (s)	Inference (μ s/obs)	Size (MB)
XGBoost	0.7683	0.7426	864.7	7.03	18.9
HistGradientBoosting	0.7643	0.7391	73.5	29.40	16.9
Random Forest	0.7398	0.7159	477.3	20.29	8703.8
Decision Tree	0.6714	0.6537	54.0	0.24	28.8
Logistic Regression	0.4685	0.4567	65.4	0.15	0.005
Linear SVM	0.4438	0.4377	4.1	0.16	0.006

Final XGBoost training required 170.34 seconds. Prediction over 10.08 million observations required 56.57 seconds, corresponding to 0.005612 ms per observation or approximately 178,000 observations per second. The serialized final model occupied approximately 6.8 MB. These values indicate that the selected classifier is computationally feasible for high-throughput offline analysis and potentially for online scoring, although end-to-end deployment latency must also include data acquisition, preprocessing, communication, and decision logic.

5 Discussion

5.1 Nonlinearity and model ranking

The ranking provides strong empirical evidence that the TEP classification boundary is not adequately represented by the evaluated linear models. XGBoost, HistGradientBoosting, and random forest substantially outperformed logistic regression and linear SVM even though all models received the same standardized predictors and run-level partitions. Tree ensembles can represent threshold effects and high-order interactions without requiring an explicit feature expansion, which is consistent with the nonlinear and feedback-driven nature of the TEP.

The difference between XGBoost and HistGradientBoosting was small: 0.0040 in mean macro-F1, with overlapping 95% confidence intervals. Thus, the defensible conclusion is that XGBoost won under the prespecified ranking rule, not that it was conclusively superior under all resampling regimes. Table 8 makes the practical consequence explicit, and it does not favour a single model. HistGradientBoosting trained 11.8 times faster than XGBoost (73.5s against 864.7s) for a deficit of 0.0040 in macro-F1 with overlapping intervals, so it dominates whenever models are retrained frequently

or the training budget is constrained. The ordering reverses at inference time, where XGBoost required 7.03 μ s per observation against 29.40 μ s, a factor of 4.2 in favour of XGBoost. Neither model dominates the other: the choice follows from whether the deployment is retraining-bound or scoring-bound, and both conclusions are invisible if only the selected model is profiled.

Model size adds a third and largely independent constraint. The random forest required approximately 8.7 GB, some 460 times the XGBoost artifact, for 0.0285 less macro-F1. Whatever its statistical merits, it is not a candidate for embedded or edge deployment, and this would not be apparent from predictive metrics alone. The two linear baselines occupied roughly 5 kB and inferred in 0.15 μ s per observation, which quantifies precisely what is being paid for the nonlinear gain of about 0.30 in macro-F1.

5.2 Why global performance is insufficient

The official macro-F1 of 0.7200 summarizes useful performance, but it does not describe a uniformly reliable 21-class diagnostic system. If only macro-F1 or accuracy had been reported, the strong results for 17 classes would obscure the near-failure of normal operation and faults 3, 9, and 15. In operational terms, low precision for normal operation and extensive exchange between normal and faulty labels can increase false alarms and missed interventions.

This result illustrates why class-level precision, recall, F1, and confusion structure should accompany aggregate measures in industrial diagnosis. It also supports a distinction between *benchmark discrimination* and *deployment readiness*. The present model is a strong benchmark classifier for most faults, but additional safeguards are necessary before it could support autonomous

decisions.

5.3 Interpretation of the ambiguity group

The identity of the difficult group is not a new finding, and it would be misleading to present it as one. Faults 3, 9 and 15 have been reported as weakly detectable since the earliest systematic studies of the TEP: monitoring statistics based on PCA, discriminant PLS and Fisher discriminant analysis report missed-detection rates close to those of nominal operation for exactly these conditions [5, 4], and later comparative studies reproduce the pattern across a wide range of data-driven statistics [18]. The contribution here is the scale and the protocol: the same group is recovered under run-level partitioning, on 10.08 million independent test observations, and it persists after the labeling artifact of Section 4.4 is removed. This strengthens the interpretation that the overlap is a property of the measured variables rather than of any particular estimator.

The mechanism, however, constrains which remedies can work. If these disturbances leave no persistent signature in the 52 measured variables, then temporal aggregation over a run reduces the variance of an estimate that is already centered on an ambiguous region: it will sharpen the decision without moving it toward the correct class. Run-level aggregation should therefore be expected to help the 17 separable classes, whose pre-onset transients are transient, far more than the group $\{0, 3, 9, 15\}$, and this asymmetry is itself a testable prediction. A hierarchical design that isolates the ambiguous group and applies a discriminator with explicit memory to classes 0, 3, 9 and 15 is the more promising direction, but its plausibility rests on whether any signature exists at all; establishing this is a question of observability, not of classifier capacity, and it is better addressed with residual-based or first-principles analysis than with additional supervised tuning.

Both proposals arise from test-set diagnosis and must therefore be evaluated under a newly defined development/test protocol; the present official test must not be reused for tuning them.

5.4 Threats to validity

Several limitations delimit the conclusions. First, TEP is a high-fidelity simulation benchmark, not a physical plant. Results may not transfer directly to sensor drift, maintenance changes, unmodeled disturbances, communication losses, or evolving operating modes in real facilities. Second, the benchmark evaluated pointwise multiclass diagnosis and did not measure detection delay, time-to-alarm, false alarms per run, or sustained alarm logic. Third, five seeds provide a useful but small basis for uncertainty estimation. The resulting t intervals characterize the implemented experiment rather than all possible data partitions.

Fourth, hyperparameter search was intentionally limited to three candidates per model family to preserve computational feasibility and transparency. A larger

search might improve individual algorithms, particularly the linear baselines or random forest. Fifth, the final model was fitted to 840,000 of the 5,250,000 available development observations, and model selection was carried out at 420,000; a ranking established at one training scale need not be invariant to scale, and the learning curve reported in Section 4.1 bounds but does not eliminate this concern. Sixth, the linear SVM was a stochastic-gradient surrogate and its result should not be read as the best attainable linear-margin performance. Seventh, the degree of parallelism was not recorded in the run manifests, so the training times of Table 8 support ordinal comparison between families, which shared one invocation, but not comparison against the timings of other stages. Eighth, interpretation focused on output behavior and did not include feature attribution or causal process analysis. Finally, the dataset had balanced class support, whereas real industrial faults are rare and highly imbalanced. Metrics, thresholds, and alarm costs would require recalibration under field prevalence.

6 Conclusion

This work presented a reproducible classical-machine-learning benchmark for 21-class fault diagnosis in the Tennessee Eastman Process. Run-level splitting, train-only standardization, separate tuning and validation, five-seed repetition, and a single frozen official test evaluation were used to reduce leakage and post-selection bias.

XGBoost achieved the highest repeated-validation macro-F1 (0.7683 ± 0.0017) and produced an official test macro-F1 of 0.7200, accuracy of 0.6754, and MCC of 0.6621 over 10.08 million observations under literal labeling, rising to a macro-F1 of 0.8054, an accuracy of 0.7949 and an MCC of 0.7852 once the pre-onset segment of each faulty run is excluded, with no refitting of any kind. The validation partition, rescored under the same protocol, moved from 0.7706 to 0.7953, so that the apparent generalization gap of +0.0483 in macro-F1 becomes -0.0125 once the comparison is made symmetric. Seventeen classes were diagnosed with F1 between 0.7429 and 0.9080, and several of them were shown to sit at the structural recall ceiling imposed by the labeling convention rather than at the limit of the classifier. Normal operation and faults 3, 9, and 15 remained strongly confounded, in agreement with two decades of TEP monitoring literature, and this confounding survives the removal of the labeling artifact.

Three conclusions follow. Nonlinear tree ensembles provide effective diagnosis for most TEP conditions. Aggregate scores overstate practical reliability when a small group of classes dominates the remaining errors. And a benchmark result on the expanded TEP dataset is not interpretable, nor comparable across studies, unless the treatment of the fault-onset window is stated explicitly. A single frozen model differs by 0.085 in macro-F1 and 0.123 in MCC between two defensible labeling conventions, which is larger than most of the improvements reported between competing architectures

on this benchmark. The effect is moreover common to every model family tested, with a relative accuracy gain between 0.142 and 0.150 across seven classifiers, so it displaces published results without reordering them only when all studies happen to adopt the same convention. We therefore recommend that the treatment of the onset window be stated as part of the experimental protocol in any observation-level study using these data.

Future work should investigate temporal aggregation, hierarchical diagnosis, probability calibration, detection delay, and cost-sensitive alarm rules under a new untouched test protocol. External validation with physical industrial data will be necessary before deployment-oriented conclusions can be drawn.

Data and Code Availability

The expanded Tennessee Eastman simulation data are publicly available from Harvard Dataverse [7]. The CSV conversion used in this study is available through Kaggle [15]. All code, configuration files, run-level split manifests, frozen hyperparameters, five-seed validation outputs, final predictions, class-level metrics, confusion matrices, SHA-256 data hashes, and the environment specification are deposited at [\[TBD: repository URL\]](#) and archived under [\[TBD: Zenodo DOI\]](#). The analysis of the fault-onset window can be reproduced from the deposited predictions without refitting any model.

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Conflict of Interest

The author declares no conflict of interest.

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